Coexisting Cloud and Snow Detection Based on a Hybrid Features Network Applied to Remote Sensing Images

Yuhan Wang, Lingjia Gu, *Member, IEEE*, Xiaofeng Li, *Member, IEEE*, Fang Gao, and Tao Jiang, *Member, IEEE*

***Abstract*— Owing to the characteristics of cloud and snow, it is difficult to detect them when they coexist. Thus, we propose an end-to-end semantic segmentation network for cloud and snow detection based on hybrid feature (CSD-HFnet) to be applied to remote sensing images (RSIs), in which cloud and snow coexist. First, the local binary pattern (LBP), gray-level co-occurrence matrix (GLCM), and superpixel segmentation are combined as basic features to preserve the textural, spatial, and shape information of the cloud and snow objects. We also propose deep learning feature extraction network to obtain the multiscale deep learning features, which is utilized to better distinguish cloud from snow. The original spectral bands, the basic features, and the multiscale deep learning features are input into the feature integration module simultaneously to form the primary multiscale hybrid features (PMHF). Then, the PMHF are filtered, sorted, and weighted by the feature filtering and sorting block and attention mechanism block to obtain the advanced multiscale hybrid features (AMHFs). Finally, the AMHFs are input into the cloud and snow segmentation network, which consists of several memory-capable gate circuits for training and validation of cloud and snow detection. The results indicate that CSD-HFnet with advanced multiscale hybrid feature (AMHF) can provide reliable detection under the condition of cloud and snow coexistence. CSD-HFnet can detect cloud and snow in multispectral RSIs of various spatial resolutions with an overall accuracy (OA) of 95.37%. Moreover, CSD-HFnet exhibits excellent cloud detection ability for red–green–blue (RGB) images with an OA of 95.88%, higher than other state-of-the-art cloud detection methods.**

***Index Terms*— Cloud and snow coexistence, cloud and snow detection, feature extraction, hybrid features network.**

I. Introduction

As a major component of the cryosphere, snow cover plays an important role in the Earth's climate system. Obtaining accurate snow cover information is therefore of great significance for climate change research as well as for the investigations of the water cycle and hydrological resources [1], [2], [3]. Optical remote sensing satellites have been used to identify snow cover, but the appearance of clouds can greatly interfere with this kind of dynamic snow monitoring [4]. With the increasing accessibility of optical data, several cloud and snow detection methods have been proposed based on optical remote sensing images (RSIs) [5], [6]. These methods can be divided into two categories: rule-based methods and deep learning-based methods [7], [8], [9], [10].

Rule-based methods detect clouds from images based on the cloud's physical properties, such as the threshold of the ground object spectrum, texture features, shape features, and so on [7]. Rule-based methods are often used to deal with a particular type of satellite imagery. Threshold rule-based methods obtain the threshold range required to detect cloud and snow from different bands based on traditional empirical values or a large number of statistical values [11], [12]. The spectral reflectance of cloud and snow is very similar in visible images, which makes them difficult to distinguish. The normalized difference snow index (NDSI) is a normalized combination of the green band and short-wave infrared band, which can detect cloud, snow, and ice from satellite images [13], [14]. The universal ratio snow index (URSI) [15] proposed by Wang et al. improves the stability of the snow index by using green band, near-infrared (NIR) band, and short-wave infrared band simultaneously. Wang et al. [16] constructed the normalized difference forest snow index (NDFSI) by using the NIR band instead of the visible band in the NDSI calculation equation in order to map the snow cover of the upper reaches of the Black River, which has forest covering the surface, and they achieved good results. Threshold rule-based methods can achieve excellent cloud and snow detection results when there is a relatively single underlying surface type, but if the underlying surface type is diverse or unevenly distributed, this can lead to misidentification of objects. Thus, threshold rule-based methods are not suitable for the detection of clouds when clouds and snow coexist [17]. To achieve excellent cloud detection performance, it is also necessary to fully consider the basic features, such as the texture and shape of clouds. Texture rule-based methods are often used for this purpose. Ground objects can be identified according to their texture features because they are different for each type of object [9].

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Yuhan Wang and Lingjia Gu are with the College of Electronic Science and Engineering, Jilin University, Changchun 130012, China (e-mail: yuhanw20@mails.jlu.edu.cn; gulingjia@jlu.edu.cn).

Xiaofeng Li and Tao Jiang are with the Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun 130102, China (e-mail: lixiaofeng@iga.ac.cn; jiangtao@iga.ac.cn).

Fang Gao is with Chang Guang Satellite Technology Company Ltd., Changchun 130000, China, and also with the College of Computer Science and Technology, Jilin University, Changchun 130012, China (e-mail: gaofang@163.com).

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Zhang et al. [18] proposed a cloud detection method that has high accuracy for high-resolution RSIs using multiple basic features, such as color, texture, and shape. Tan et al. [19] proposed a method that can accurately detect clouds by combining threshold and basic features (spectral, texture, and superpixel features). Li et al. [7] developed a method that integrates multiple features, first performing threshold cloud recognition based on spectral features and then using texture features to improve the cloud detection results before generating the final cloud mask. Although the overall accuracy (OA) of rule-based methods is high, when the spectral reflectivity of the noncloud area is high, the area is often misidentified as cloud [20]. Since most of the cloud detection methods proposed only use original spectral and basic features, there is still room to improve cloud detection accuracy by using higher level features [7].

In recent years, deep learning methods have been widely applied in cloud feature extraction and cloud detection by using convolution kernels of different sizes to extract deep and high-level abstract features from input images [21], [22]. Recently, image semantic segmentation techniques based on deep learning have achieved amazing performance. As introduced in [23], [24], and [25], these deep learning-based semantic segmentation algorithms have achieved significant improvements for distinguishing cloud pixels, further improving cloud detection accuracy. In the cloud and snow coexistence environment, the labels in the cloud and snow detection are mainly divided into cloud, snow, and others. Wieland et al. [26] proposed a method to detect clouds based on a modified U-Net (MU-Net). Chai et al. [27] proposed the modified SegNet (MSegNet) based on a deep convolutional neural network (CNN) to detect clouds and shadows in Landsat imagery. MU-Net and MSegNet are the typical deep learning networks with an encoder–decoder structure, but cloud objects are often omitted. Effective semantic segmentation network methods, such as Deeplab V3+ [28], PSPNet [29], and so on, are also employed for cloud detection [30]. Although the cloud detection methods based on Deeplab V3+ and PSPNet can achieve superior cloud detection results, there are several minor drawbacks, such as poor performance when applied to small areas [24], [25]. Yang et al. [24] used a CNN-based method, named CDnetV1, to detect clouds for high-resolution RSIs. However, if the basic features of cloud, such as shape and texture, are ignored, cloud detection algorithms may not be able to fully guarantee the stability and accuracy of cloud detection results even using deep learning-based methods. To improve the accuracy of cloud detection, Xie et al. [31] apply both simple linear iterative clustering (SLIC) and CNN model in the cloud detection task, which can more comprehensively utilize the cloud features to obtain the cloud detection results. However, this method does not consider the scenario of cloud and snow coexistence. Guo et al. [25] proposed CDnetV2 for cloud and snow detection, which uses a feature pyramid module to extract multiscale contextual information without the loss of resolution and coverage and boundary refinement blocks to refine object boundaries, but the drawback is that a large number of parameters need to be set. At present, most deep learning-based methods mainly detect only clouds from images, and there are few that detect snow.

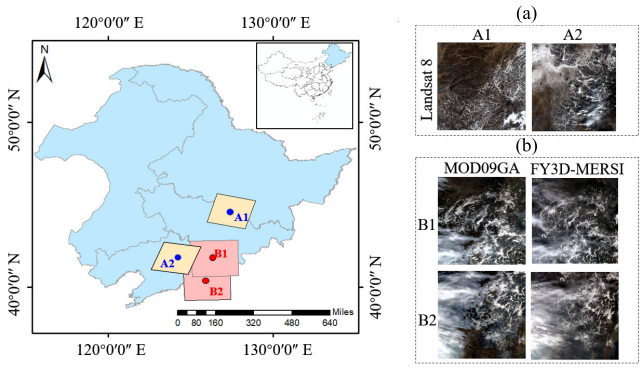


Fig. 1. Map and sample photographs of the experimental areas.

Since both clouds and snow show high reflectivity and have complex and irregular texture feature problems, it can cause confusion between clouds and snow in the detection of cloud and snow. Thus, the accurate identification of cloud and snow has been a challenging task in the process of cloud and snow detection. To address the above problems, we propose an end-to-end cloud and snow detection method based on hybrid feature network (CSD-HFnet), and the main contributions of this article are as follows.

1) The proposed multiscale and multifeature module can comprehensively extract spectral features, basic features, and multiscale deep learning features of experimental data and further compose them as the primary multiscale hybrid features (PMHFs).

2) The proposed feature integration module can utilize feature filtering and sorting block and attention mechanism block to filter, sort, and assign weights for PMHF to obtain advanced multiscale hybrid features (AMHF) of cloud and snow.

3) By combining the AMHF with the cloud and snow segmentation network, which consists of four gate circuits, accurate cloud and snow detection results that multispectral images can be obtained. The proposed method has good robustness and great practicality not only in multispectral images and red–green–blue (RGB) images with different spatial resolutions but also in continuous multitemporal data with dynamic cloud and snow detection results, which is of great significance to realize snow detection and snowmelt runoff analysis.

The remainder of this article is organized as follows. Section II introduces the physical experimental areas and data. In Section III, we introduce the proposed CSD-HFnet framework. The cloud and snow detection results from the images of various spatial resolutions are described in Section IV. Section V presents a discussion of the results, and Section VI gives the conclusions.

II. Experimental Areas and Datasets

A. Experimental Areas

The experimental areas are mainly located in northeast China, as shown in Fig. 1. The climate of the experimental

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TABLE I
CLOUD AND SNOW DETECTION DATA INFORMATION

Experiment	Case	Type	Data	Date	Band	Spatial /Time Resolution
Moderate-to-high Spatial Resolution Image	A1/A2	Experimental data	Landsat 8	01/27/2019 12/17/2018	R, G, B, NIR, SW1, SW2	30 m/16 day
	A1	GT	GF	01/26/2019-01/28/2019	R, G, B	1 m/1day
	A2	GT	GF	12/16/2018-12/20/2018	R, G, B	1 m/1day
Moderate Spatial Resolution Image	B1/B2	Experimental data	FY3D MERSI	02/12/2019	R, G, B, NIR	250 m/1 day
	B1/B2		MOD09GA	02/12/2019	R, G, B, NIR, SW1, SW2	500 m/1 day
	B1/B2	GT	Landsat 8	02/12/2019	R, G, B, NIR, SW1, SW2	30 m/16 day
RGB High Spatial Resolution Cloud Detection Datasets	C	Experimental data	GF dataset	unknown	R, G, B	1 m/4 day
			ZY-3 dataset			2 m/5 day
		GT	GF dataset			1 m/4 day
			ZY-3 dataset			2 m/5 day

areas is characterized by temperate continental semi-humid monsoon patterns, with warm and rainy summers and extensive snowfall in winter. They are considered stable snow areas with an average snow depth of over 5 cm throughout the snow season. We conduct three experiments on the proposed method. Experiment A considers moderate-to-high spatial resolution and multispectral RSIs covering areas A1 and A2. Experiment B considers moderate spatial resolution and multispectral RSIs covering areas B1 and B2. Experiment C considers the public cloud detection datasets based on RGB with high spatial resolution, including the Gaofen (GF) dataset and ZiYuan-3 (ZY-3) dataset.

B. Experimental Data

The experimental data are divided into the experimental cloud and snow detection dataset and the ground truth (GT) dataset, as shown in Table I.

1) Experimental Cloud and Snow Detection Dataset: As listed in Table I, the experimental cloud and snow detection dataset includes the images of moderate-to-high spatial-resolution from the Landsat 8 Operational Land Imager (OLI) [32], images of moderate spatial-resolution from the Fengyun-3D (FY3D) satellite [33] and MOD09GA of the Moderate Resolution Imaging Spectroradiometer (MODIS) [34], and RGB high spatial resolution cloud detection datasets from the GF dataset [35] and ZY-3 dataset [36].

a) Moderate-to-high spatial-resolution images: The revisit period of the Landsat 8 OLI satellite is 16 days. Landsat 8 has eight multispectral bands with a spatial resolution of 30 m and a panchromatic band with a spatial resolution of 15 m [32]. We use six multispectral bands: red, green, blue, NIR, and short-wave infrared (SW1 and SW2).

b) Moderate spatial-resolution images: The Fengyun-3D (FY3D) satellite is a Chinese second-generation polar-orbiting meteorological satellite. The Moderate Resolution Spectral Imager II (MERSI-II) is one of the main payloads of the FY3D satellite and is equipped with 25 channels. Among the 25 channels, there are six with a spatial resolution of 250 m and 19 with a spatial resolution of 1 km [33]. We use the red, green, blue, and NIR bands with a spatial resolution of 250 m from FY3D MERSI as the experimental bands.

MODIS provides daily global observations at spatial resolutions of 250 m to 1 km. The MODIS-level two gridded product (MOD09GA) has six multispectral bands (red, green, blue, NIR, SW1, and SW2) with a spatial resolution of 500 m [34].

c) RGB high spatial resolution cloud detection datasets: The GF dataset [35] is a cloud detection dataset produced by the Beihang University of China. The dataset contains 2443 RGB images with a spatial resolution of 1 m taken by the GF series of satellites [37]. The GF dataset contains the images of different environments, where clouds and snow are present, both separately and coexisting. The image size is 3000×2000.

The ZY-3 dataset [36] is a set of ZY-3 satellite images. The ZY-3 satellite is China's first civilian high-resolution optical transmission stereomapping satellite. Its optical sensor can obtain images at a resolution of 2 m [38]. The ZY-3 dataset consists of 432 images of scenes, including 232 snow-covered scenes and 200 snow-free scenes.

2) GT Dataset: To discern the real cloud and snow distribution in Experiments A and B, the RSIs with higher resolution are used as the GT data. The RSIs of the GF series satellites are used as the GT data for Experiment A, and the Landsat 8 images with a higher resolution than MOD09GA and FY3D are used as the GT data for Experiment B. For the RGB high spatial-resolution images, the GF dataset and ZY-3 dataset provide the corresponding GT cloud images.

III. METHODOLOGY

This article proposed an effective CSD-HFnet under the condition of cloud and snow coexistence. The framework of the proposed method is shown in Fig. 2, which mainly includes a multiscale and multifeature module, a feature integration module, and a cloud and snow segmentation network. The module of multiscale and multifeature contains three branches to obtain different features from the input images. For the first branch, the original spectral features are directly introduced into the next feature integration module. For the second branch, the multiscale deep learning features of cloud and snow are obtained by using the deep learning feature extraction network. For the third branch, the basic features of the image, such as texture features, spatial features, and shape features of cloud and snow, are obtained by gray scale co-generation

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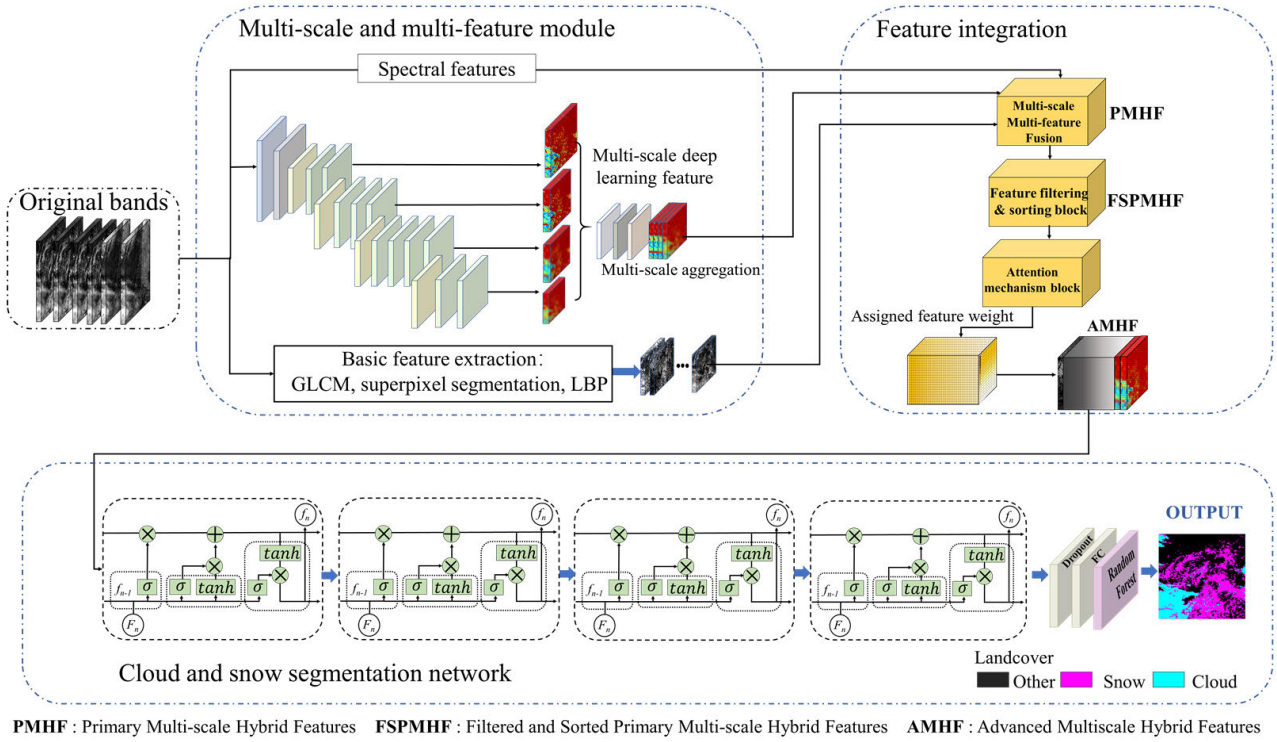


Fig. 2. Flowchart of the proposed CSD-HFnet.

matrix (GLCM), local binary pattern (LBP), and superpixel segmentation, respectively. Furthermore, the original spectral features, multiscale deep learning features, and basic features are input into the feature integration module simultaneously. Through the processing of the multiscale and multifeature fusion, all the input features are composed into the PMHF. Then, the PMHFs are filtered, sorted, and weighted by the feature filtering and sorting block and attention mechanism block, respectively, to obtain the AMHF. Finally, the AMHFs are input into the cloud and snow segmentation network, which consists of several memory-capable gate circuits for training and validation of cloud and snow detection. The trained CSD-HFnet model is applied to detect cloud and snow in multispectral and RGB images with different spatial resolutions even in the case that cloud and snow coexist, and the dynamic cloud and snow detection is carried out for continuous multitemporal images. All the experiments are carried out on Windows 10 operating system and are run within NVIDIA GeForce RTX 3090 Ti.

A. Preprocessing

Before carrying out research on cloud and snow detection, RSI preprocessing, such as radiometric correction, atmospheric correction, and so on, should first be carried out on the experimental data, and then, the raw spectral bands of the remote sensing data are obtained after preprocessing.

B. Feature Extraction

1) Basic Features Extraction:

a) GLCM texture features: GLCM [39] is often used to deal with the spatial relationship of image pixel values. Texture features can comprehensively reflect the gray level statistics, spatial distribution, and structural information of the image and can also describe the repetition or change between image pixels [40]. The GLCM texture feature can analyze the grayscale information contained in the image and calculates the frequency of each bit of grayscale information. The advantage of GLCM texture features is that they can be used to accurately predict and reflect comprehensive information about the direction, adjacent intervals, and variation range of the image's gray level. GLCM texture features are widely used in cloud detection [41]. Since the texture of cloud is usually smooth, while the texture of snow and other ground objects is usually rough, thus clouds have different GLCM texture features from snow and other ground objects, so the use of GLCM texture features is beneficial to distinguish cloud and snow and improve the accuracy of cloud and snow detection.

b) LBP features: Not only GLCM but also LBP features are widely used in cloud detection [41], [42]. LBP can describe the local spatial structure of the image. It has the characteristics of rotation invariance, good robustness to illumination changes, and extreme sensitivity to noise [43]. The LBP operator takes the difference between the gray values of a central pixel and its eight adjacent pixels. If an adjacent pixel has a value greater than that of the center pixel, the value of the pixel is marked as 1; otherwise, it is marked as 0. Then, matrix encoding is performed to convert the 8-bit binary number into a decimal number as the LBP value of the center pixel. The LBP features contain the spatial information of cloud and snow, and the difference of spatial information of cloud and snow can be used to help the detection of cloud and snow. LBP can be calculated using the following

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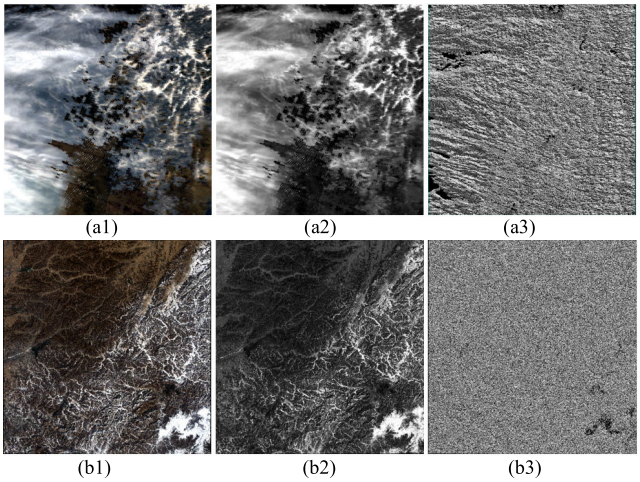


Fig. 3. LBP texture feature. (a1) Original. (a2) Gray. (a3) LBP feature. (b1) Original. (b2) Gray. (b3) LBP feature.

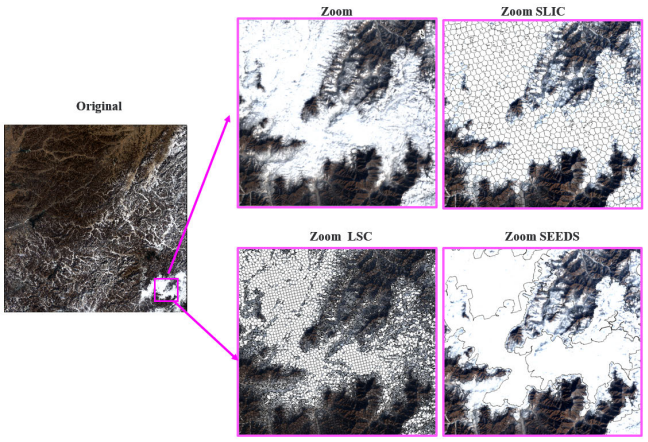


Fig. 4. Shape features for superpixel segmentation.

equation [43]:

$$\mathrm{LBP}_{N,R}(x_c, y_c) = \sum_{n=0}^{N-1} s(h_n - h_C) \times 2^n$$
$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{1}$$

where (x_c, y_c) represents the center pixel and h_C represents the gray value of the center pixel. Taking the pixel point (x_c, y_c) as the center of a circle, the circle area with radius R is defined as the neighborhood of the center pixel point, N represents the number of pixel points in the field, and h_N is the gray value of the neighboring pixel point. Fig. 3 shows the LBP features of a given RSI.

c) Shape Features for Superpixel Segmentation: Compared with common segmentation methods that only use image pixels as the basic unit, superpixel segmentation not only divides an image by pixel but also clusters segmented pixels with similar characteristics. Representative superpixel segmentation algorithms include SLIC [44], linear spectral clustering (LSC) [45], and superpixels extracted via energy-driven sampling (SEEDS) [46]. Superpixels greatly reduce the computational complexity of image processing and improve computational efficiency.

SLIC adapts a k-means clustering approach to efficiently generate superpixels, which requires that the pixels sorted into the same superpixel be as similar as possible. The principle of the LSC algorithm is to map the pixel data of an image to a high-dimensional space and then cluster it. The LSC algorithm has good robustness to noise and can effectively utilize the features of image color, brightness, and textural structure to avoid oversegmentation and generate highly compact feature boundaries. The SEEDS algorithm divides the image into a regular grid, and the grid is the initial superpixel. The edge of the current superpixel is then gradually moved, such that the pixels inside the superpixel have as uniform color characteristics as possible in order to quickly complete the segmentation. Fig. 4 shows the superpixel segmentation results. Compared with other methods, it can be seen that the SEEDS method can effectively retain the details of cloud and snow shape features in the image without destroying the structure of the image. Therefore, we select the segmentation results of SEEDS as the basic shape features of an image.

2) Deep Learning Feature Extraction: The deep learning feature extraction network is an important part of CSD-HFnet that provides important features for cloud and snow detection. The structure and parameters of the deep learning feature extraction network are shown in Fig. 5. The deep learning feature extraction network is composed of multiple CONV BLOCK modules and IDENTITY BLOCK modules. The purpose of the IDENTITY BLOCK module is to deepen the network and achieve deep feature extraction. The purpose of the CONV BLOCK module is to change the size and the number of channels in the images. The first part of this network consists of a convolutional layer, a BN layer, a RELU layer, and a Maxpool concatenated to compress the image. The subsequent parts of this network are four combinations of CONV BLOCK and IDENTITY BLOCK, each of which can output one deep learning feature, and here, four multiscale ($N/2 \times N/2$, $N/4 \times N/4$, $N/8 \times N/8$, and $N/16 \times N/16$) deep learning features can be obtained. Then, the multiscale deep learning features are adapted to be consistent with the input image size ($N \times N$) through the 2-D average pooling layer and the fully connected layer. After experiment, we set each iteration with batch size 10, dropout 0.5, learning rate 0.001, and iteration epoch 1000 in feature extraction. To clearly observe the process that the multiscale deep learning features extracted by the deep feature extraction network can increase the distinction between cloud and snow, this article visualizes the multiscale deep learning features using the heatmap module. Heatmap visualization is a method of graphically representing numerical data in the form of hot and cold spots using a warm to cold color scheme. The heatmap module describes the important parts of the feature map by visualizing multiscale deep learning features using Grad-Cam [47], [48], [49], allowing the network to focus on the target region information. Different features in the heatmap have different levels of highlighting, which can clearly show the differentiation of feature characteristics. Warm colors indicate data points with high probability of other

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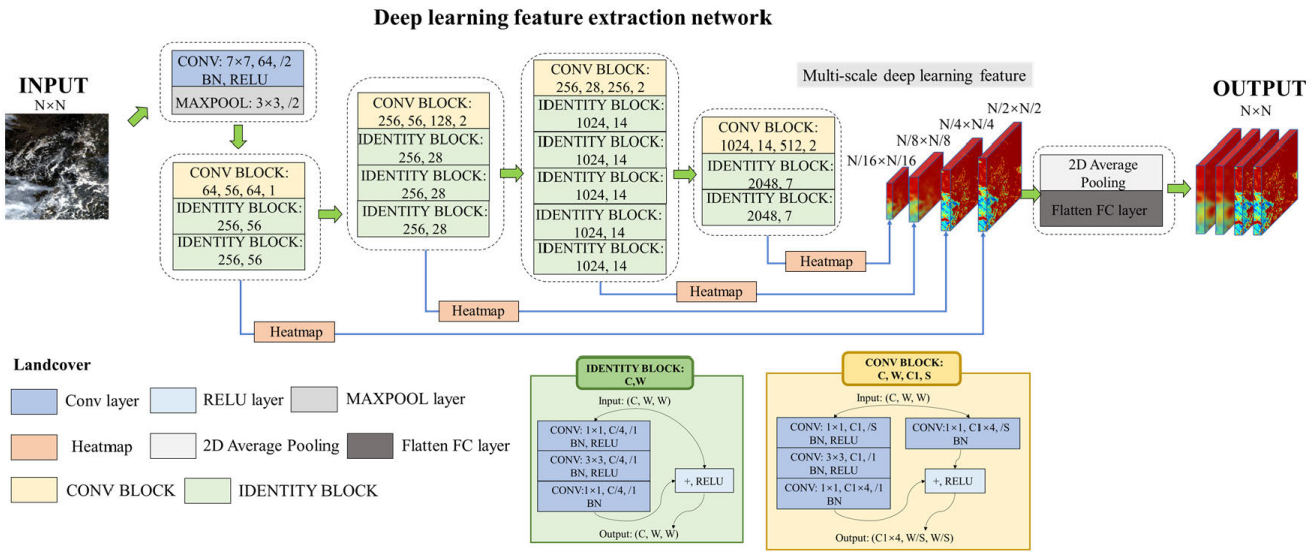


Fig. 5. Structure of the deep feature extraction network.

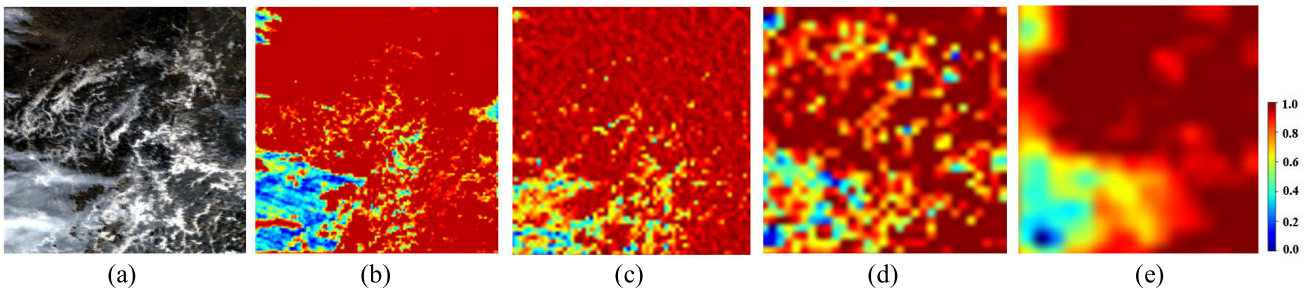


Fig. 6. Multiscale deep learning feature from the deep feature extraction network. (a) $N \times N$ original image. (b) $N/2 \times N/2$ feature. (c) $N/4 \times N/4$ feature. (d) $N/8 \times N/8$ feature. (e) $N/16 \times N/16$ feature.

ground objects, intermediate colors indicate data points with high probability of snow, and the cool colors indicate data points with high probability of clouds.

To highlight the importance that using multiscale deep learning features can increase the recognition between cloud and snow, this article shows the extracted deep learning features in the form of the heatmap, and it can be found that the feature differences among cloud, snow, and other features are more obvious in the multiscale deep learning features, which are beneficial to cloud and snow detection. The extracted multiscale deep learning features are stretched to be consistent with the original image, as shown in Fig. 6, and it can be found that the multiscale deep learning features have different scales and sharpness due to the depth of networks. The closer value of the high-brightness area in the heatmap is to 1 (red), the greater probability of other ground objects is at that position. Conversely, the closer value of the low-brightness region of the heatmap is to the region of 0 (blue), the greater probability of cloud is at that position. When the value of a point on the heatmap is close to the median values of 0–1 (yellow or green), the greater probability that position is covered with snow.

C. Feature Integration

To utilize the extracted features effectively, CSD-HFnet uses the feature integration module to fuse and process the features. The steps of feature integration for all input features are described as follows.

D. Cloud and Snow Segmentation

The segmentation network with gate circuit structure is selected as the main framework for cloud and snow detection. It has the ability to remember information and is particularly sensitive to temporal changes and feature changes [50], which is more conducive to daily cloud and snow detection with high accuracy for continuous multitemporal data.

Fig. 7 shows the detailed structure of the cloud and snow segmentation network. Considering that the number of gates affects the accuracy of the detection results, the cloud and snow segmentation network mainly consists of four gate circuit modules [51], [52], [53]. The architecture of each gate module includes n gate cells, where n denotes the number of features contained in the AMHF, and a fully connected layer that improves generalizability is also connected with the output of the last gate module. The final layer of the cloud and snow segmentation network is an RF classifier including three neurons corresponding to the probabilities of three classes (cloud, snow, and others), which is the output layer of the model. The gate circuit unit extracts the high-level nonlinear information available in the AMHF supplied to the RF classifier to detect cloud and snow. After experiment, we set each iteration with

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Algorithm 1 Algorithm for Feature Integration

Input: Original spectral features, basic features, multi-scale deep learning features, label.

Output: Advanced hybrid feature combination

1. Multiple multi-scale feature fusion: The original spectral features, basic features and multi-scale deep learning features are composed to the Primary Multi-Scale Hybrid Features (PMHF) as $\{F_1, F_2, \ldots, F_M\}$.
2. Calculate Pearson correlation coefficients for PMHF and label (L). The result constitutes the set $P = \{P_{L-F1}, P_{L-F2}, \ldots, P_{L-FM}\}$.

$$P_{A-B} = \frac{\sum_{i=1}^{n}(A_i - \overline{A})(B_i - \overline{B})}{\sqrt{\sum_{i=1}^{n}(A_i - \overline{A})^2}\sqrt{\sum_{i=1}^{n}(B_i - \overline{B})^2}}$$

where A and B are the characteristic variables, $\overline{A}$ and $\overline{B}$ are the mean values of the characteristic variables, and n is the number of values taken by the variables.

3. The corresponding P-value features less than 0 are removed, and the remaining PMHF are sorted according to the P-value to obtain the Filtered and Sorted Primary Multi-scale Hybrid Features (FSPMHF).
4. Using the self-attention mechanism of the attention module, different feature weights are assigned to FSPMHF to adaptively generate Advanced Multi-scale Hybrid Features (AMHF) that can focus on the important features for cloud and snow detection.

Note: M is the number of original input features.

batch size 10, dropout 0.5, learning rate 0.001, and iteration epoch 10 000 in the cloud and snow detection training.

E. Quantitative Evaluation Index

The detection results were quantitatively evaluated using six indicators: precision [54], recall [55], OA [56], kappa [57], F1 score [58], and the mean intersection-over-union (mIOU) [59], as in (2)–(7), as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$OA = \frac{TP + TN}{TP + FN + TN + FP} \tag{4}$$

$$p_e = \frac{(TP + FN) \times (TP + FP) + (FP + TN) \times (FN + TN)}{(TP + TN + FP + FN) \times (TP + TN + FP + FN)}$$

$$\text{and Kappa} = \frac{OA - p_e}{1 - p_e} \tag{5}$$

where true positive (TP), false positive (FP), and false negative (FN) represent the number of correctly extracted, incorrectly extracted, and incorrectly not extracted pixels, respectively.

TABLE II
QUALITY EVALUATION (%) FROM MODERATE-TO-HIGH SPATIAL RESOLUTION RSIS

Area	OA	Kappa	mIOU	PA	Recall	F1
A1	94.75	85.51	82.21	92.51	87.89	90.01
A2	95.37	88.01	82.93	93.07	88.19	90.38
Mean	95.06	86.76	82.57	92.79	88.04	90.20

The F1 score and mIOU are two comprehensive evaluation indicators. The F1-score is a representation of the harmonic mean of precision and recall, which is calculated by

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

The mIOU is applied to characterize accuracy at the segment level and is calculated by

$$\text{IOU} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall} - \text{Precision} \times \text{Recall}}$$

$$\text{and mIOU} = \sum_{i=0}^{k} \text{IOU}_i \Big/ k \tag{7}$$

where k is the total number of segmentation patches.

IV. EXPERIMENTAL RESULTS

A. Cloud and Snow Detection for Moderate-to-High Spatial Resolution RSIs

The results of cloud and snow detection for moderate-to-high spatial resolution RSIs for Experiment A are shown in Fig. 8. It can be seen that there are cloud occlusions in the original Landsat 8 images of A1 and A2, which included thick clouds, thin clouds, and broken clouds. In the case of cloud and snow coexistence, it is easy to confuse cloud and snow, but the proposed method can detect clouds from experimental areas A1 and A2 under this condition. The precision analysis is shown in Table II. Cloud and snow detection using CSD-HFnet obtained detection results with an OA of 94.75% and 95.37% in experimental areas A1 and A2, respectively. This implies that the proposed method is able to perform cloud and snow detection for moderate-to-high resolution multispectral RSIs.

B. Cloud and Snow Detection for Moderate Spatial Resolution RSIs

The data from experimental areas B1 and B2 were captured by MODIS and FY3D at the same geographical location and on the same date. Using experimental data with different spatial resolutions for the same geographical location can not only test the performance of the proposed method at cloud and snow detection but they can also be used as validation data for each other. Fig. 9 shows the cloud and snow detection results for moderate resolution RSIs.

Differences in the clouds between the two experimental areas can be observed. In B1, there exist thick clouds in the lower left corner of the MOD09GA image and the cloud coverage area has expanded into thin clouds in the FY3D image. In the B2 area, the cloud area of MOD09GA has increased compared with FY3D. According to all evaluation

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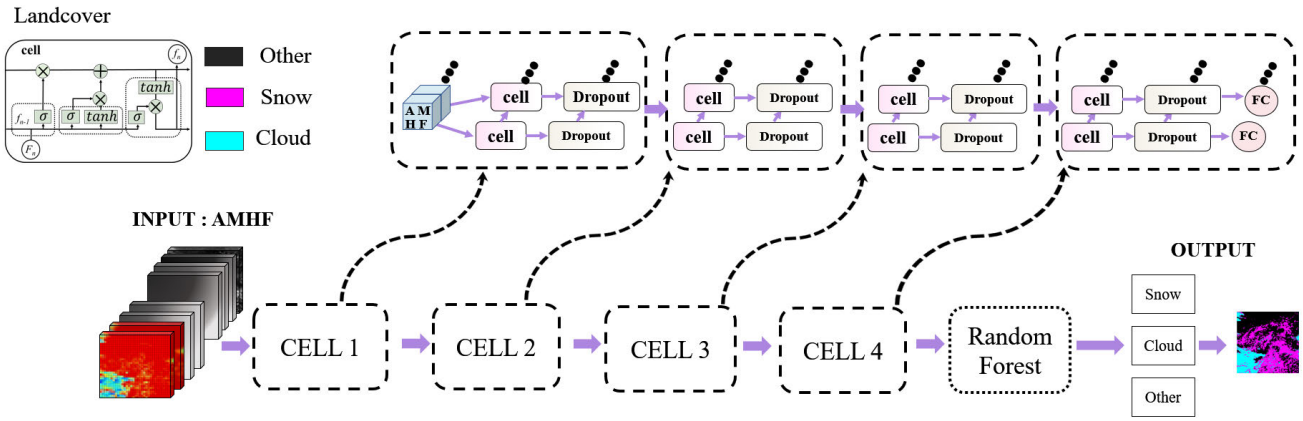


Fig. 7. Structure of the cloud and snow segmentation network.

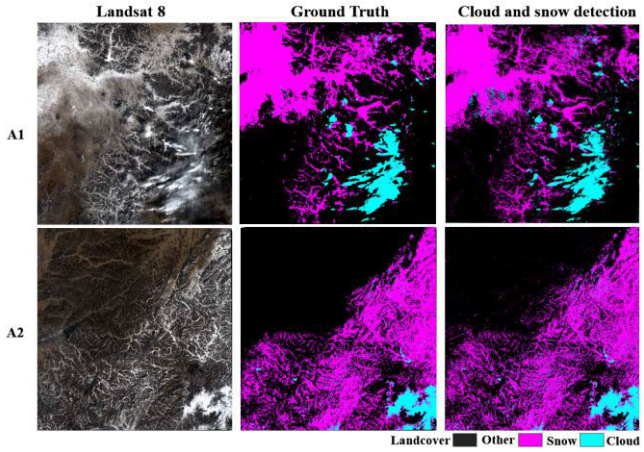


Fig. 8. Detection of cloud and snow from moderate-to-high spatial resolution RSIs.

TABLE III
QUALITY EVALUATION (%) FROM MODERATE SPATIAL RESOLUTION RSIS

Data	MOD09GA			FY3D		
Area	B1	B2	Mean	B1	B2	Mean
OA	92.37	92.08	92.23	92.49	93.28	92.89
Kappa	87.20	87.12	87.16	88.63	89.68	89.16
mIOU	84.18	82.89	83.54	85.93	86.27	86.10
PA	92.36	91.21	91.79	92.38	92.84	92.61
Recall	90.47	89.68	90.08	92.51	92.20	92.36
F1	91.33	90.35	90.84	92.42	92.48	92.45

indicators, the detection results of FY3D are more reliable than those of MOD09GA because the latter has a lower spatial resolution, resulting in MOD09GA appearing to show more snow pixels than FY3D. To verify the ability of the proposed method to identify cloud and snow at different spatial resolutions, this article uses RSIs from Landsat 8 images of the same date as the GT data. The precision analysis is described in Table III.

It can be seen that CSD-HFnet can achieve excellent cloud and snow detection results for multispectral MOD09GA images, with OA reaching 92.37%, a kappa coefficient up to 87.20, and mIOU up to 84.18. The cloud and snow detection results for the multispectral FY3D images showed an OA up to 93.28%, a kappa coefficient up to 89.68, and an mIOU up to 86.27. It is concluded that this method can effectively distinguish cloud from snow in moderate-resolution RSIs.

TABLE IV
QUALITY EVALUATION (%) FROM RGB HIGH SPATIAL RESOLUTION CLOUD AND SNOW DETECTION DATASETS

Dataset	OA	Kappa	mIOU	PA	Recall	F1
GF dataset	94.59	87.97	83.97	92.80	89.67	91.10
ZY dataset	94.62	88.09	86.75	93.43	92.41	92.82
Mean	94.61	88.03	85.36	93.11	91.04	91.96

C. RGB High Spatial Resolution Cloud Detection Datasets: GF and ZY-3 Datasets

The images of the GF dataset and ZY-3 dataset contain red, green, and blue bands of 1 and 2 m, respectively. The images in which cloud and snow coexist are used as experimental data to verify whether the proposed method is also effective when applied to RGB images. The results of cloud and snow detection performed on the high spatial resolution cloud detection datasets are shown in Fig. 10.

It can be seen that the cloud and snow objects in the RGB images are extremely similar and visually indistinguishable when they coexist. Nevertheless, the position of clouds can be accurately located using the proposed method, and snow can be distinguished from other ground objects. The precision analysis is carried out, as shown in Table IV. It can be seen that the characteristics of the GF and ZY-3 datasets with extremely high spatial resolution result in more detailed information, which can further improve the accuracy of cloud and snow detection.

D. Dynamic Cloud and Snow Detection Experiment

In this article, a dynamic cloud and snow detection experiment was conducted in the experimental area A1 for 16 continuous days from December 11, 2018 to December 26, 2018, and the obtained dynamic cloud and snow detection results are shown in Fig. 11. The changing states of cloud and snow for 16 days can be observed in Fig. 11, where the experimental area A1 had a coexistence of cloud and snow in the results of

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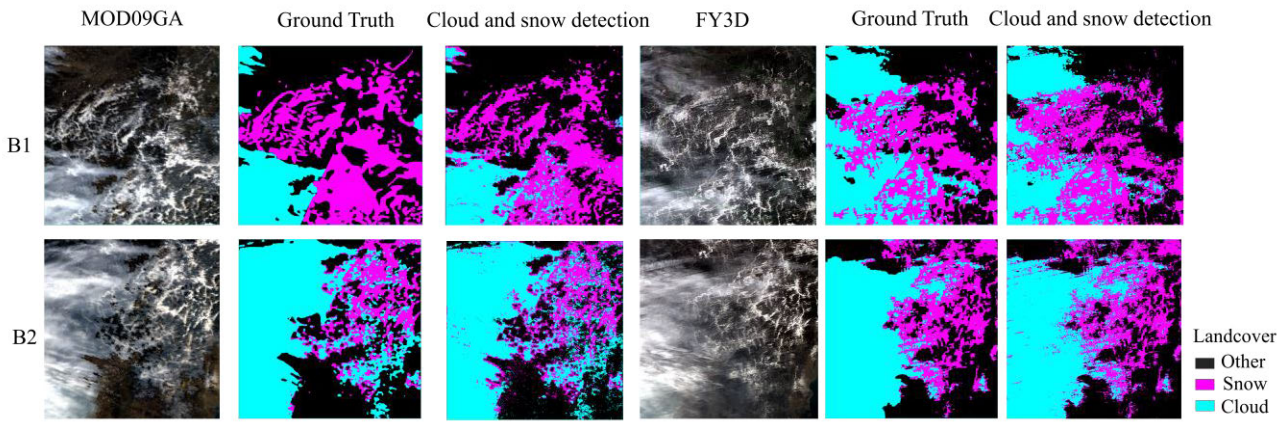


Fig. 9. Detection of cloud and snow for moderate spatial resolution RSIs.

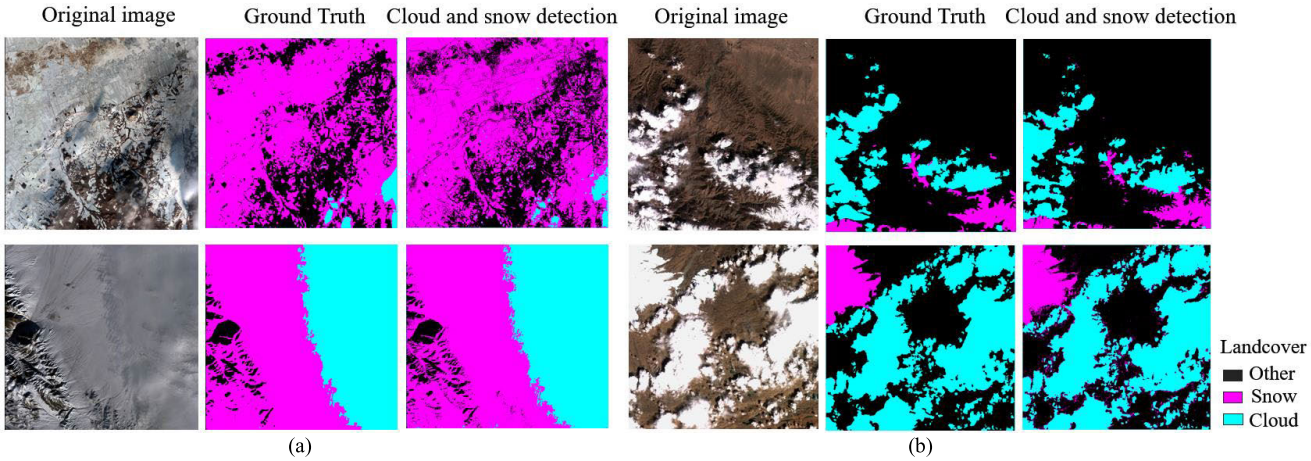


Fig. 10. Detection of cloud and snow for RGB high spatial resolution cloud and snow detection datasets. (a) GF dataset. (b) ZY-3 dataset.

T_1, T_2, and T_6–T_{11}, and a small snowfall and a large snowfall occurred in T_3 and T_{12}, respectively. The experimental results indicate that the CSD-HFnet has good detection performance in continuous multitemporal images.

V. Discussion

A. Analysis of Cloud and Snow Detection With Different Feature Combinations

In order to evaluate the effect of using the feature extraction ability of the cloud and snow segmentation network based on gated circuits and different hybrid features combinations on the cloud and snow detection results, we conducted four comparative experiments. The first and second columns of Fig. 12 are the original and GT images, respectively. The third column displays the results obtained from the selected data using spectral features, which have serious cloud and snow confusion problems. The fourth column displays the results obtained from the selected data using spectral features and basic features, demonstrating significant confusion between cloud and snow objects. The fifth column displays the result obtained from the selected data using spectral features and multiscale deep learning features, which shows that a large number of ground objects in the Landsat 8 images have been misclassified as snow, and the snow objects of the MOD09GA, FY3D, and RGB cloud detection dataset images are incompletely extracted. The sixth column of Fig. 12 shows the cloud and snow detection results obtained using AMHF, which is obviously superior to the other feature combinations.

The cloud and snow segmentation network also has a certain feature extraction capability, and the third column of Fig. 12 shows the cloud and snow detection results obtained using the feature extraction capability of the cloud and snow segmentation network. The accuracy results of the various feature combination are shown in Fig. 13. The highest accuracy was achieved by using AMHF, which suggests that using AMHF is particularly important for ensuring the accuracy of cloud and snow identification.

B. Analysis of Cloud and Snow Detection for RGB Images With Different Spatial Resolutions

Most high spatial resolution datasets only provide R, G, and B bands. Therefore, to verify the effectiveness of the proposed method applied to RGB images that lack NIR and short-wave infrared bands, this section discusses the detection performance for RGB images with various spatial resolutions. The detection results are shown in Fig. 14. The first to third rows of Fig. 14 show the original images, the GT images, and the extraction results, respectively. The first to firth columns

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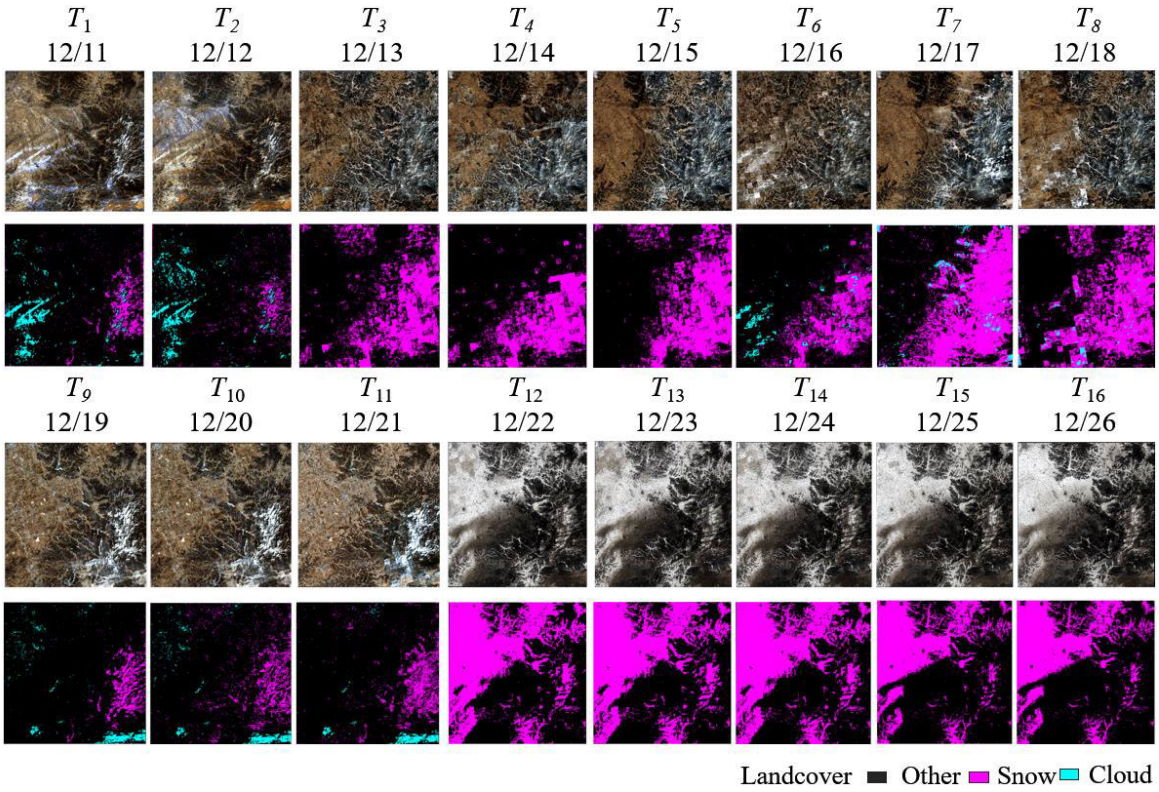


Fig. 11. Result of the dynamic cloud and snow detection in 2018.

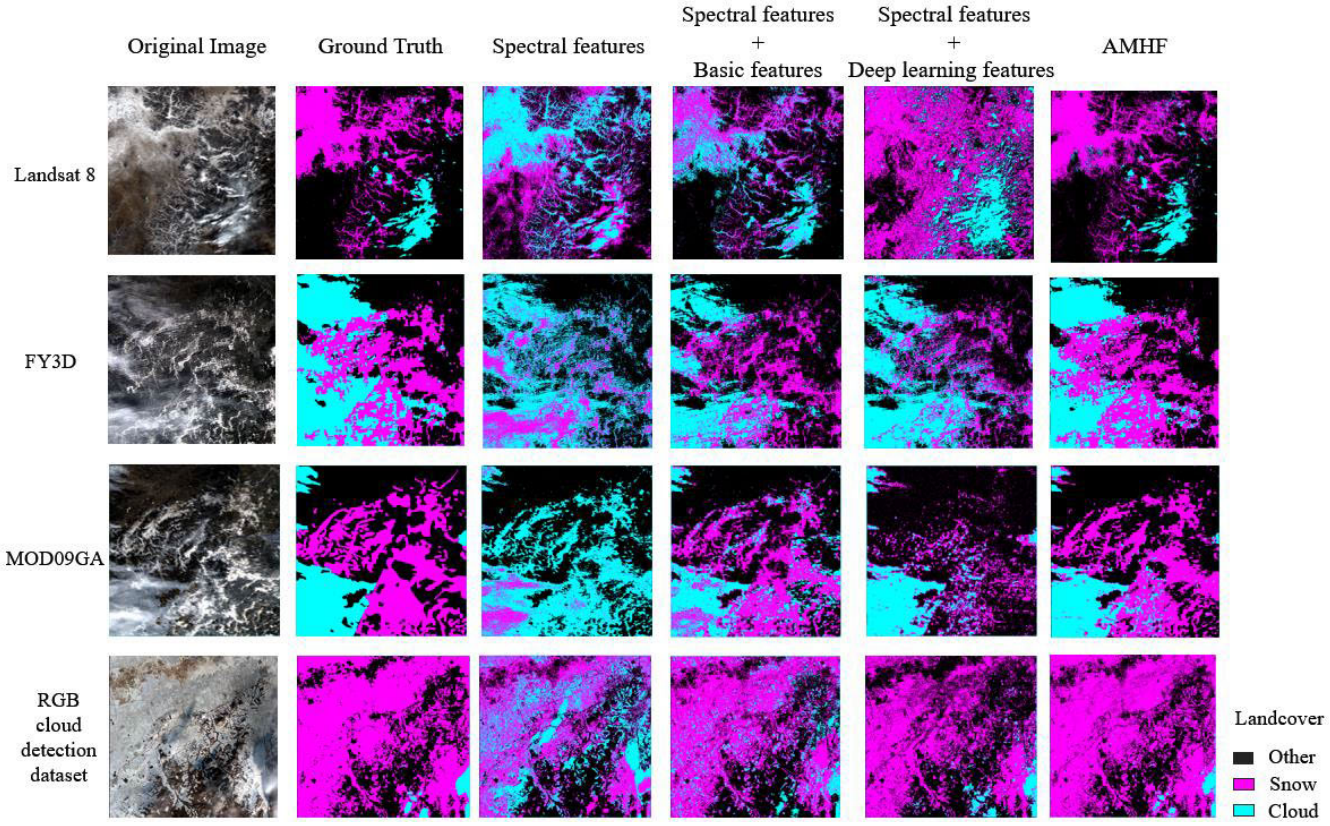


Fig. 12. Detection of cloud and snow from different feature combinations on multispectral images.

are the detection results from MOD09GA, FY3D, Landsat 8, GF, and ZY-3 with RGB bands. From Fig. 14, it can be found that the proposed method still has excellent cloud and snow detection performance in RGB images with different spatial

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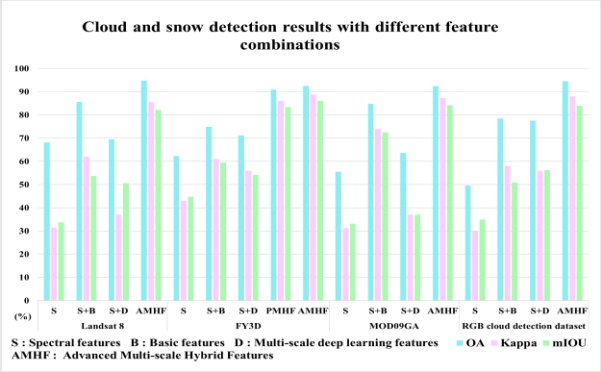


Fig. 13. Quality evaluation from different feature combinations on multispectral images.

TABLE V
MEAN QUALITY EVALUATION (%) FROM RGB IMAGES WITH DIFFERENT SPATIAL RESOLUTIONS

Method	OA	Kappa	mIOU	PA	Recall	F1
MOD09GA	86.22	78.73	75.46	84.69	87.40	84.44
FY3D	87.09	80.60	77.14	86.97	87.55	87.09
Landsat 8	90.63	82.73	80.98	88.24	89.89	89.33
GF	94.59	87.97	83.97	92.80	89.67	91.10
ZY-3	94.62	88.09	86.75	93.43	92.41	92.82

resolutions, and the cloud and snow distribution states in the obtained detection results are consistent with the GT images.

The accuracy of the cloud and snow detection for RGB images with different spatial resolutions is shown in Table V. It is observed that the cloud and snow detection results with relatively high accuracy for the RGB images of various spatial resolutions can be obtained using CSD-HFnet when the NIR, SW1, and SW2 bands are not present. Moreover, according to all evaluation indicators, the detection OA for high spatial-resolution images is slightly better than that for the corresponding low spatial-resolution images because the former contains more details. It can also be observed that the detection accuracy for multispectral images is better than that for the corresponding RGB image, indicating that multispectral images provide more spectral information than RGB images, thereby improving the accuracy of cloud and snow detection.

C. Comparative Experiment of Cloud and Snow Detection Based on Snow Indices

The NDSI, NDFSI, and URSI are the representative snow identification methods used in traditional cloud and snow detection studies for MODIS RSIs. The equations of NDSI, NDSFI, and URSI are [14], [15], [16]:

$$\text{NDSI} = \frac{b_3 - b_6}{b_3 + b_6} \tag{8}$$

$$\text{NDFSI} = \frac{b_5 - b_6}{b_5 + b_6} \tag{9}$$

$$\text{URSI} = \frac{b_3}{b_3 + b_6} \tag{10}$$

where b_3, b_5, and b_6 are the green, NIR band, and short-wave infrared band, respectively.

TABLE VI
MEAN QUALITY EVALUATION (%) OF NDSI, NDFSI, URSI, AND THE PROPOSED METHOD

Method	OA	Kappa	mIOU	PA	Recall	F1
NDSI	68.81	51.54	40.25	69.67	68.21	40.25
NDFSI	50.23	27.66	32.45	55.08	62.51	32.44
URSI	70.81	52.01	47.99	68.05	68.86	47.99
CSD-HFnet	92.23	87.16	83.54	91.79	90.08	90.84

The NDFSI index uses the NIR band to replace the green band in the NDSI equation, and the URSI utilizes both green band, NIR band, and short-wave infrared band. To verify the snow identification ability of NDSI, NDFSI, and URSI under the condition of cloud and snow coexistence, the proposed method is compared with these three methods, and the comparison results are shown in Fig. 15. It can be seen that a large amount of cloud and snow were misidentified using the NDSI-based method, whereas the NDFSI-based method extracted more nonsnow objects and fewer cloud objects than was present in the GT images. Due to the poor image quality of the NIR band of the MOD09GA data, the NDFSI results for experimental areas B1 and B2 have stripe noise. The cloud and snow detection results obtained based on URSI are better than those of NDSI and NDFSI, but there are also some cloud and snow confusion cases in the URSI results. However, the distribution states of cloud and snow in the detection results obtained by CSD-HFnet are consistent with the GT images. An analysis of the mean quality evaluations of the proposed method, NDSI-based method, NDFSI-based method, and URSI-based method is shown in Table VI for experimental areas B1 and B2.

The proposed method is much more accurate than the NDSI-based, NDFSI-based, and URSI-based methods. When cloud and snow did not coexist in an image, setting the threshold based on NDSI, NDFSI, or URSI may have advantages for identification. However, when there is coexistence of cloud and snow, the NDSI-based, NDFSI-based, and URSI-based methods cannot achieve satisfactory identification results and readily misclassify cloud and snow objects. The proposed method, on the other hand, is a deep learning cloud and snow detection method that comprehensively considers the basic features and deep learning features of clouds, snow, and other ground objects.

D. Analysis of Cloud and Snow Detection by SOTA Deep Learning Methods

CDnetV2 [25], CDnetV1 [24], PSPNet [29], DeeplabV3+ [28], MSegNet [27], and MU-Net [26] have been considered as competing SOTA methods. These methods achieve promising cloud detection results when applied to the ZY-3 dataset, and the proposed method is compared with them in terms of cloud detection from the ZY-3 dataset to verify its cloud detection ability. The cloud detection results of the SOTA deep learning methods are shown in Fig. 16. The ZY-3 dataset contains images and GT images, as shown in the first and second columns of Fig. 16. It can be seen that MSegNet, MU-Net, and PSPNet extracted fewer cloud objects than those in the GT images.

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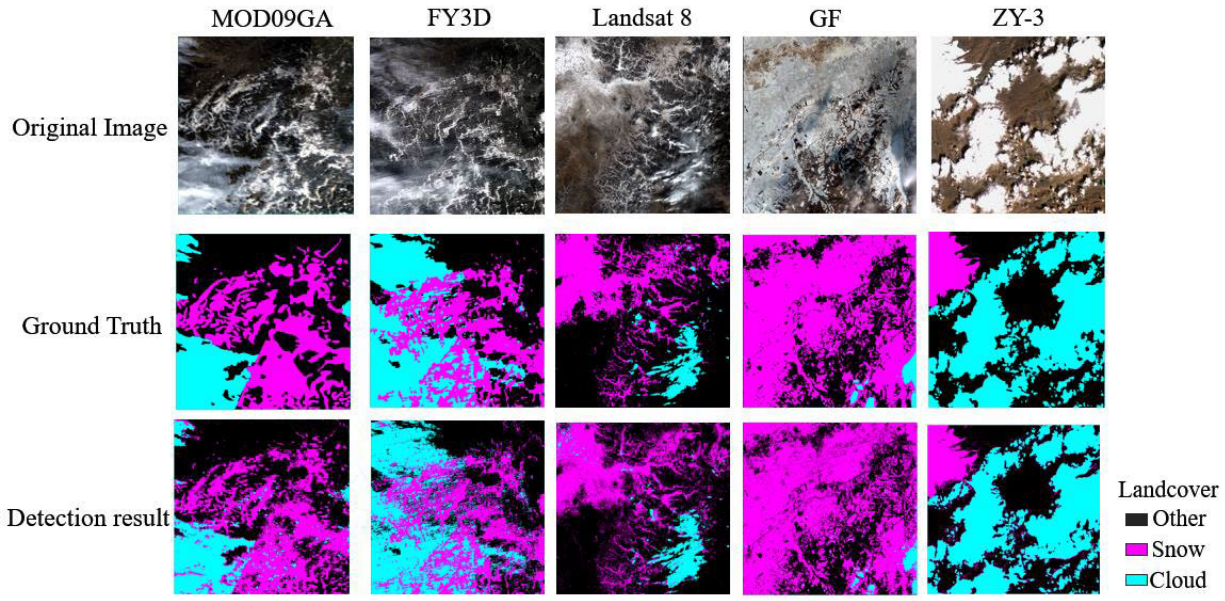


Fig. 14. Detection of cloud and snow for RGB images with different spatial resolutions.

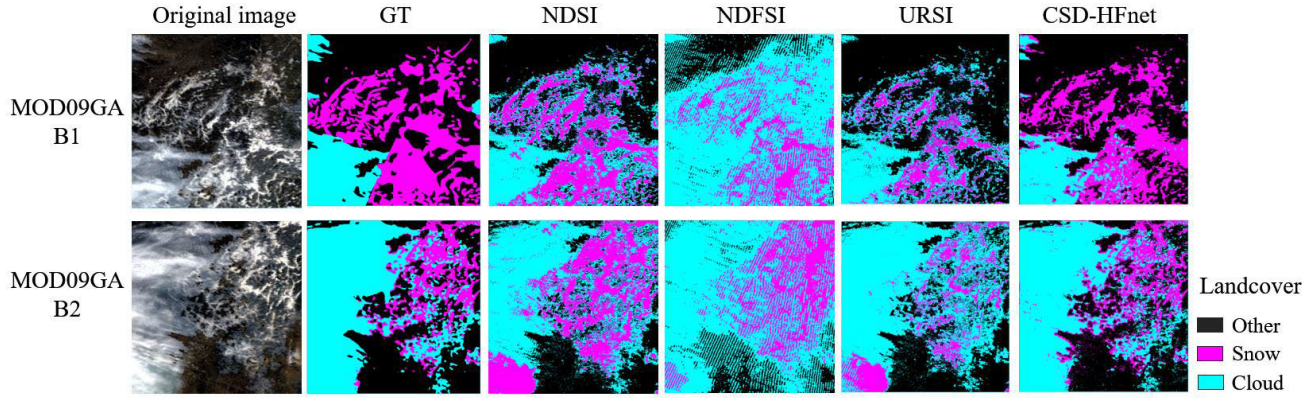


Fig. 15. Comparison of NDSI, NDFSI, URSI, and the proposed method.

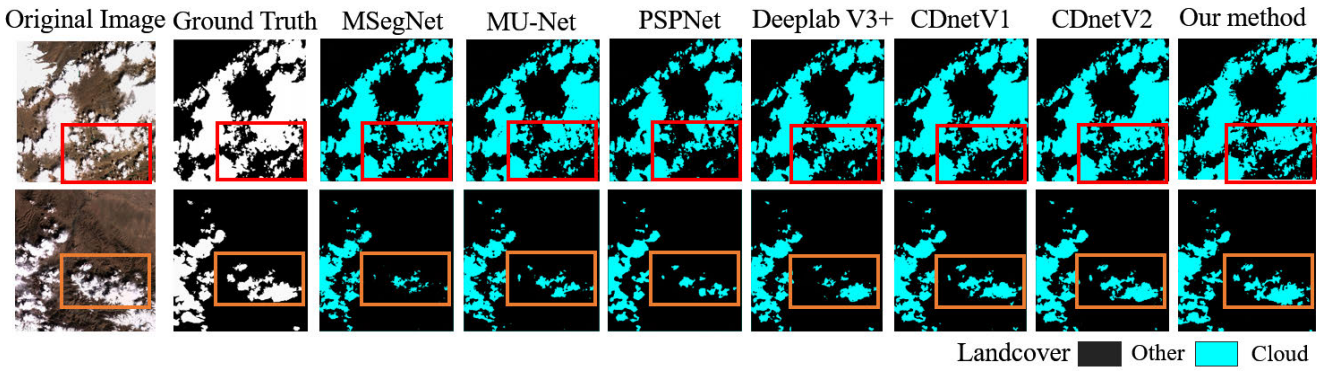


Fig. 16. Cloud and snow detection from SOTA deep learning methods.

The cloud detection performance of DeeplabV3+ was more superior to that of MSegNet, MU-Net, and PSPNet, but it still had shortcomings. Although CDnetV1 and CDnetV2 obtained very similar cloud distributions as the GT, there were still more omissions of cloud in the red and orange boxes compared with our proposed method.

This article analyzes the average evaluation metrics of cloud detection results and evaluates the computational complexity of these SOTA methods using two evaluation criterions, that is, the number of params and running time. All the experiments are carried out on Windows 10 operating system and run within NVIDIA GeForce RTX 3090 Ti. Table VII shows the

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TABLE VII
PERFORMANCE AND EFFICIENCY EVALUATION FROM SOTA DEEP LEARNING METHODS

Method	Performances (%)						Efficiency	
	OA	Kappa	mIOU	PA	Recall	F1	Params (M)	Running times (s)
MSegNet [28]	90.86	75.57	81.2	73.78	86.13	79.47	29.79	1.28
MU-Net [27]	91.62	76.7	82.51	74.44	87.39	80.39	8.60	1.09
PSPNet [30]	90.58	75.36	81.63	76.02	87.52	81.36	46.63	1.05
DeeplabV3+ [29]	91.8	77.65	82.62	75.3	87.76	81.05	40.33	1.14
CDnetV1 [25]	93.15	79.21	82.8	82.37	86.72	84.48	64.86	1.26
CDnetV2 [26]	95.76	82.51	86.62	87.75	88.58	88.16	65.91	1.31
CSD-HFnet [Our]	95.88	87.47	87.14	87.86	93.13	90.41	25.98	1.52

performance and efficiency analysis of the proposed method and SOTA method. It can be seen from Table VII that the number of params for the proposed method is less than MSegNet, PSPNet, DeeplabV3+, CDnetV1, and CDnetV2, suggesting that the model has relatively high efficiency. Although the running times of all compared methods are close, the proposed method performed better than the other competing methods for all the evaluation indices, which demonstrates its superiority.

VI. CONCLUSION

It is a challenging task to obtain accurate cloud and snow detection in the cloud and snow coexistence areas due to their high similar characteristics, such as color and texture. Moreover, the spatial resolution and band information of RSIs also affects the detection accuracy. To solve these problems, we proposed a deep learning method, named CSD-HFnet, for cloud and snow detection in RSIs under the condition of cloud and snow coexistence. The proposed method focuses on fusing the original spectral bands, basic features, and the multiscale deep learning features to form AMHF to improve cloud and snow detection accuracy. The basic features, which obtained through GLCM, LBP, and SEEDS, can provide the basic information of images. The CSD-HFnet contains two important deep learning networks. One is a deep learning feature extraction network for extracting the multiscale deep learning features to further distinguish clouds and snow. The other is the cloud and snow segmentation network, which consists of several memory-capable gate circuits for training and validation of cloud and snow detection to be applied to detect cloud and snow in RSIs with different spatial resolutions. Excellent results were achieved for RSIs with spatial resolutions ranging from 1 to 500 m, and the highest obtained an OA of 95.37%. It is not only effective when applied to multispectral images but also to RGB images of various spatial resolutions. In the future, we will apply the proposed method to detect cloud and snow in other satellite images. In addition, more consideration is needed for thin cloud and snow detection.

ACKNOWLEDGMENT

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Yuhan Wang received the M.Sc. degree from the College of Electronic Science and Engineering, Jilin University, Changchun, China, in 2020, where she is currently pursuing the Ph.D. degree.

Her work focuses on snow monitoring by remote sensing data and spatiotemporal fusion. Her research interests include remote sensing data processing and deep learning algorithm research.



Lingjia Gu (Member, IEEE) received the Ph.D. degree in electrical circuits and systems from Jilin University, Changchun, China, in 2008.

She is currently a Professor with the College of Electronic Science and Engineering, Jilin University. From 2010 to 2012, she held a post-doctoral research position with the Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun. Her research interests include remote sensing data processing and digital image processing.



Xiaofeng Li (Member, IEEE) received the B.Sc. and M.Sc. degrees in mathematics from Jilin University, Changchun, China, in 2002 and 2005, respectively, and the Ph.D. degree in remote sensing and geosciences from the Chinese Academy of Sciences, Beijing, China, in 2008.

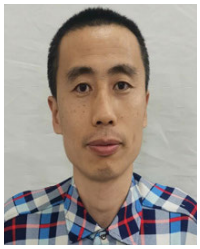
He is currently a Professor with the Research Center of Remote Sensing and Geoscience, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun. His research interests include snow parameters inversion by remote sensing data, image processing, and agriculture remote sensing.

Dr. Li is also the Chair of IEEE GRSS Changchun Chapter.



Fang Gao received the B.Sc. and Ph.D. degrees from Jilin University, Changchun, China, in 2010 and 2016, respectively.

He is currently a Senior Engineer with the Key Laboratory of Satellite Remote Sensing Application Technology of Jilin Province, Jilin University, and with Chang Guang Satellite Technology Company Ltd., Changchun. His research interests include remote sensing image processing and artificial intelligence.



Tao Jiang (Member, IEEE) received the Ph.D. degree in cartography and geographic information system from the Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun, China, in 2019.

He is currently an Engineer with the Microwave Remote Sensing Group, Research Center of Remote Sensing and Geoscience, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences. His research interests include microwave radiometer design and the development and detection of radio frequency interference.

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